PHY202 Practice Exam on chapter(s): 23, 24

Name:____

These questions assess your understanding of the material from Chapter(s) 23, 24 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

Note that I provide references on practice exams so you can find additional help if you cannot solve a problem; I do not give this information on in-class exams. The related chapter and section are provided after each question. E.g. "(Chapter 18, Section (5) Electric Field Lines - cp1e_ex18_4)" means that the question was inspired by OpenStax College Physics First Edition's Chapter 18, Example 4, which is found within Chapter 18, Section 5. See the end of this practice exam for a list of the sources.

1. Calculate the motional emf induced along a $31.\text{-km}$-long conducting object moving at an orbital speed of $7.8\,\frac{\text{km}}{\text{s}}$ perpendicular to Earth's $31.\text{-}\mu\text{T}$ magnetic field.
(Chapter 23, Section (3) Motional EMF - cp1e_ch23_ex2)

ANSWER: ______________________________

2. A portable x-ray unit has a step-up transformer, the 130 V input of which is transformed to the $1.9\times10^{5}\text{-V}$ output needed by the x-ray tube. The primary has $14.$ loops and draws a current of 1.9 A when in use. Find the current output of the secondary.
(Chapter 23, Section (7) Transformers - cp1e_ch23_ex5)

ANSWER: ______________________________

3. A generator coil is rotated 1.5 revolutions in 6.2 ms. The 150-turn circular coil has a 2.6-cm radius and is in a uniform 0.60-T magnetic field. What is the maximum emf that is induced?
(Chapter 23, Section (5) Electric Generators - cp1e_ch23_ex4)

ANSWER: ______________________________

4. How much energy is stored in a 1.3-mH inductor when a $36.\text{-A}$ current flows through it?
(Chapter 23, Section (9) Inductance - cp1e_ch23_ex8)

ANSWER: ______________________________

5. Calculate the self-inductance of a 0.15 m-long, 0.022 m-diameter solenoid that has 240 coils.
(Chapter 23, Section (9) Inductance - cp1e_ch23_ex7)

ANSWER: ______________________________

6. Suppose a $80.$-turn coil lies in the plane of the page in a uniform magnetic field that is directed into the page. The coil originally has an area of 0.69 m^2. It is then stretched for 0.47 s until it has no area. What is the magnitude of the induced emf if the uniform magnetic field has a strength of 0.55 T?
(Chapter 23, Section (2) Faraday's Law of Induction: Lenz's Law - cp1e_ch23_p8)

ANSWER: ______________________________

7. A battery charger meant for a series connection of several nickel-cadmium batteries needs to have a $17.\text{-V}$ output to charge the batteries. It uses a step-down transformer with a 150-loop primary and a 110-V input. If the charging current is $16.\text{ A}$, what is the input current?
(Chapter 23, Section (7) Transformers - cp1e_ch23_ex6)

ANSWER: ______________________________

8. On its highest power setting, a 730-W microwave oven projects microwaves onto a 0.44-m-by-0.44-m area. Calculate the peak electric-field strength in the electromagnetic waves emitted by the microwave oven.
(Chapter 24, Section (4) Energy in Electromagnetic Waves - cp1e_ch24_ex4)

ANSWER: ____________________________

9. Suppose a 0.68-H inductor is in series with a $0.65\text{-}\Omega$ resistor and a current of $13.\,\text{A}$ flows through them. Find the current that flows through the inductor and resistor $1.7\,\text{s}$ after the battery is disconnected.
(Chapter 23, Section (10) RL Circuits - cp1e_ch23_ex9b)

ANSWER: ____________________________

10. A generator coil is rotated $0.81\,\text{revolutions}$ in $23.\,\text{ms}$. The $92.$-turn circular coil has a 3.0-cm radius and is in a uniform 1.2-T magnetic field. What is the average emf that is induced?
(Chapter 23, Section (5) Electric Generators - cp1e_ch23_ex3)

ANSWER: ____________________________

11. What is the characteristic time constant for a 2.8-mH inductor in series with a $5.1\text{-}\Omega$ resistor?
(Chapter 23, Section (10) RL Circuits - cp1e_ch23_ex9)

ANSWER: ____________________________

12. What is the maximum strength of the magnetic field in an electromagnetic wave that has a maximum electric-field strength of $420\,\text{V/m}$?
(Chapter 24, Section (2) Production of Elecromagnetic Waves - cp1e_ch24_ex1)

ANSWER: ____________________________

cp1e: College Physics (1st Edition) by OpenStax
PSE_Serway_4e: Physics for Scientists and Engineers (4th Edition) by Serway
tkd: I did not refer to anything while writing this question